PCB DESIGN PROJECT REPORT

POWER SUPPLY CONDITIONING BOARD

Assignment Question 13

Submitted by

VEDANT SAROHA

Designed using KiCad

1. Project Overview

The objective of this project is to design a Power Supply Conditioning Board that converts an external DC input into a regulated 5 V supply. The design uses an L7805 linear voltage regulator, input and output capacitors for filtering and stability, and an LED indicator with a series resistor to indicate the presence of the output supply.

2. Main Components and Working

• **J1 – DC input jack:** Provides the external 9 V DC input.

• **U1 – L7805 regulator:** Regulates the input voltage to a nominal 5 V output.

• **C1 – 330 nF and C3 – 10 µF:** Input-side capacitors used for supply bypassing and smoothing.

• **C2 – 100 nF and C4 – 10 µF:** Output-side capacitors used for bypassing and smoothing.

• **R1 – 1 kΩ and D1 – LED:** The resistor limits LED current; the LED provides a visual power indication.

• **J2 – 5 V output terminal:** Provides the regulated output connection to the external load.

The DC input is applied at J1. The L7805 regulator generates the regulated 5 V rail. The capacitors help reduce input and output ripple and support stable operation. The LED branch indicates that the output rail is present. The regulated supply is available at J2.

3. Circuit Schematic

The schematic shows the 9 V DC input, L7805 regulator, filtering capacitors, LED indicator branch and 5 V output connector.

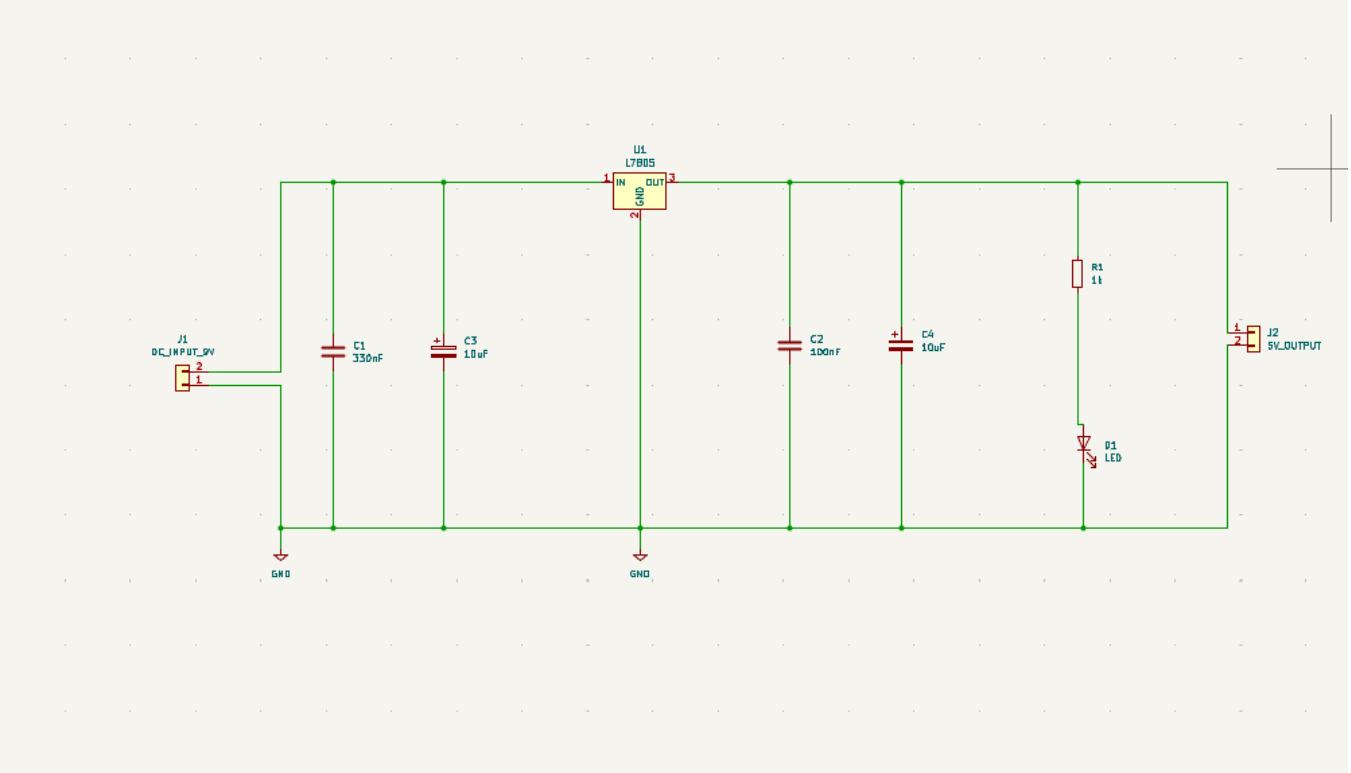


Figure 1. Power Supply Conditioning Board circuit schematic.

4. Routed PCB Layout (Footprints and Tracks)

The PCB layout shows the assigned component footprints and completed copper routing between the pads.

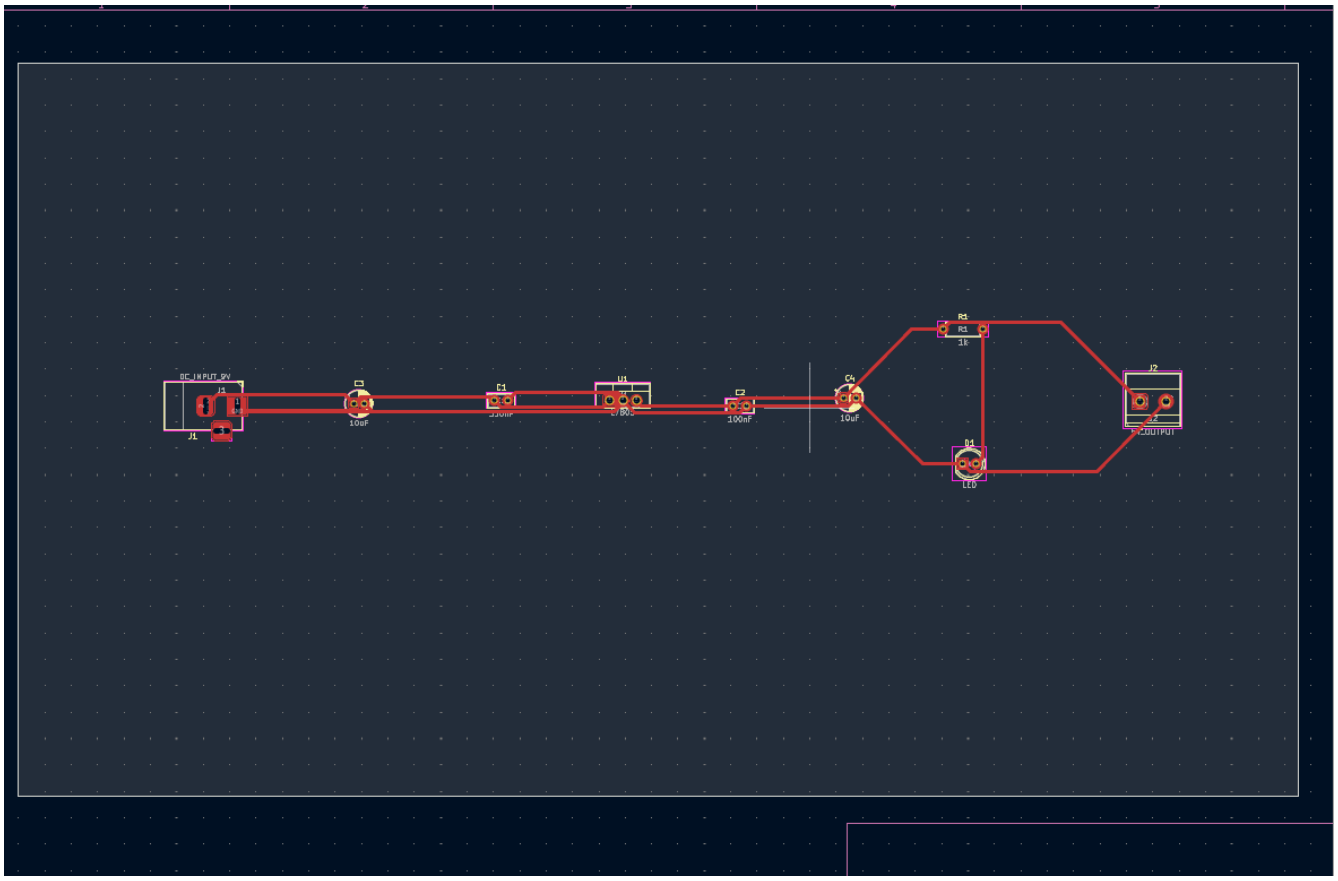


Figure 2. Two-dimensional PCB layout with component footprints and routed tracks.

5. 3D PCB View

The 3D model provides a visual inspection of component placement and the approximate physical arrangement on the board.

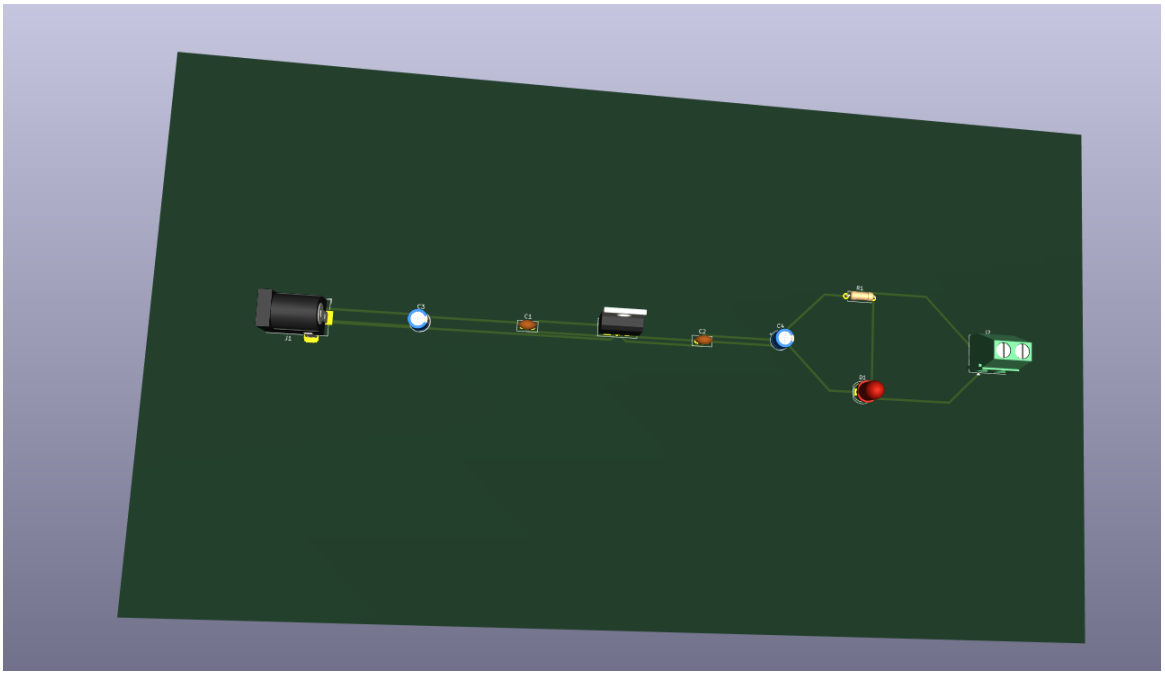


Figure 3. 3D PCB view – first viewing angle.

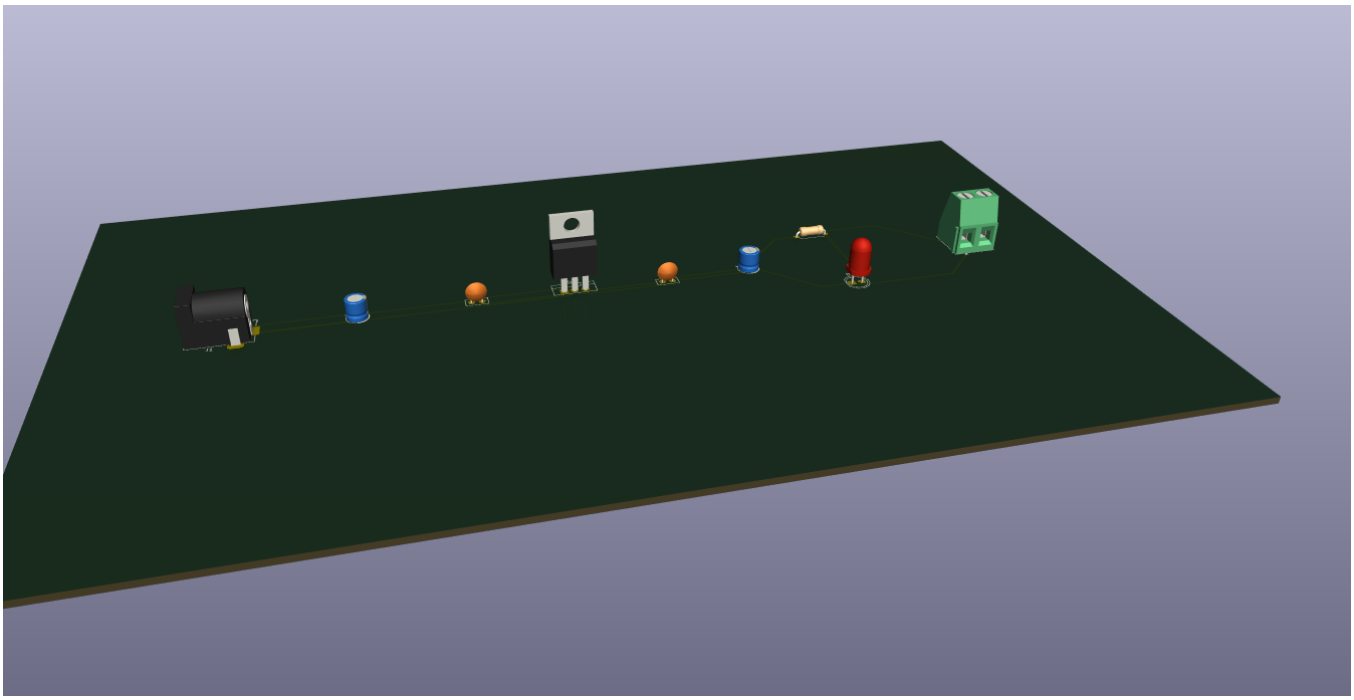


Figure 4. 3D PCB view – alternate viewing angle.

6. Bill of Materials (BOM)

The following table is transcribed from the supplied KiCad Bill of Materials export.

ID	Designator	Footprint	Qty.	Value / Description
1	C1	C_Disc_D5.0mm_W2.5mm_P2.50mm	1	330nF
2	C2	C_Disc_D5.0mm_W2.5mm_P2.50mm	1	100nF
3	C3	CP_Radial_D5.0mm_P2.00mm	1	10uF
4	C4	CP_Radial_D5.0mm_P2.50mm	1	10uF
5	D1	LED_D5.0mm	1	LED
6	J1	BarrelJack_Horizontal	1	DC_INPUT_9V
7	J2	TerminalBlock_Phoenix_MKDS-1,5-2-5.08_1x02_P5.08mm_Horizontal	1	5V_OUTPUT
8	R1	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1	1k
9	U1	TO-220-3_Vertical	1	L7805

7. Design Output Summary

The project deliverables shown in this report include the circuit schematic, routed PCB layout, 3D PCB views and the component Bill of Materials. The Gerber and drill output files were also generated separately in KiCad and should be included in the project repository for fabrication review.

Note: This report documents the design screenshots and BOM supplied for preparation. Electrical performance, thermal behaviour and operation under a real load should be verified with suitable measurements before manufacturing or use.